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https://www.biodiversity-exploratories.de/

Monitoring Climate Impacts on Germany’s Biodiversity

Final Project Presentation
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Project Motivation

Background in behavioural ecology

- How the environment impacts individual animals

Step back and look at large-scale, system-level processes

- Biodiversity and climate
- Big data and machine learning

Biodiversity

The variety of life on earth
(plants, animals, fungi, bacteria, etc.)

Healthy biodiversity is essential for functioning ecosystems and human societies

Healthy biodiversity is hard to define - Stability is important!

Can be impacted by various environmental features

Our changing world

Climate change is disrupting weather patterns

Extreme weather events are becoming more frequent:

- Heatwaves
- Droughts
- Floods

Impacts on biodiversity

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A Machine Learning model that predicts regional patterns of biodiversity stability in Germany in response to extreme weather conditions

Accurate predictions are important:

- Early warning system (before permanent damage/ecosystem collapse occurs)
- Inform policy decisions

The Data Science bit

Weather Data

- OpenMeteo Historic Weather API
- Daily weather data for each Bundesland between 2004-2024

- Feature engineering
  - Monthly weather
  - Drought, heatwave and flood risks
  - Lagged features

The Data Science bit

Biodiversity Data

- GBIF—the Global Biodiversity Information Facility
- ‘Citizen science’ observations geotagged in Germany 2004-2024
  - > 30M observations

- Feature engineering
  - Monthly, regional aggregation
  - Observation effort
  - Biodiversity stability (Turnover - Jaccard index)

The Data Science bit

Model

XGBoost Regressor

- predict Biodiversity (in)stability per Bundesland per month based on weather data

The Data Science bit

Results

Intuition model R^2: 4% explained variance

Test set R^2: 43% explained variance

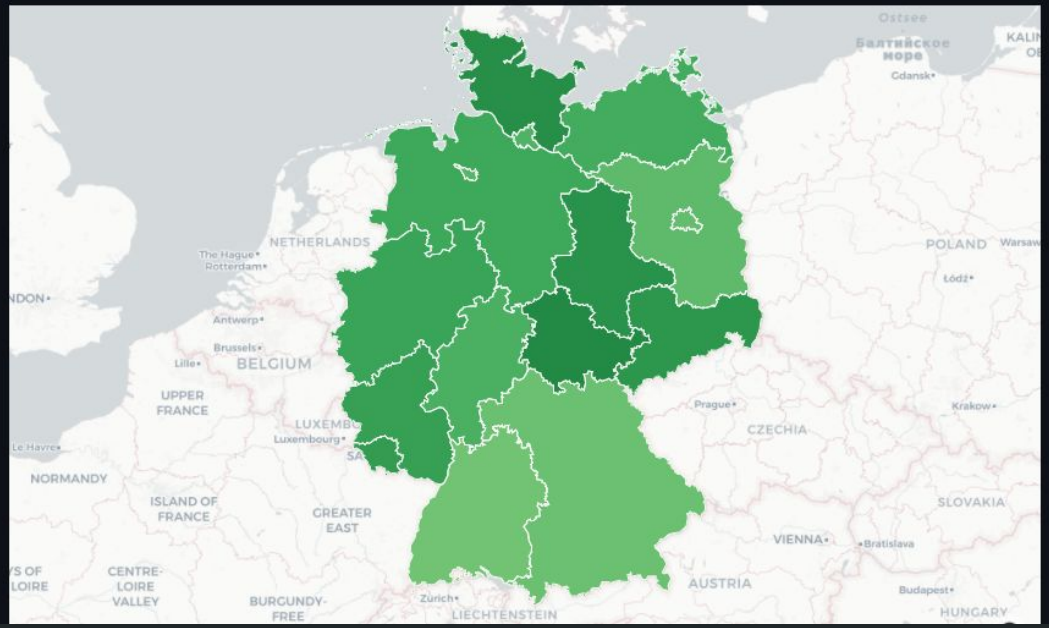
Pretty good for noisy ecological data!

- Weather is only one of many drivers of biodiversity stability
- Low resolution of weather and biodiversity data

WildSignal Biodiversity Forecast

This chart shows how ecosystem dynamics are expected to change relative to recent conditions under different IPCC climate change projections. Higher values indicate more ecological change (not necessarily better biodiversity), while lower values indicate more stable ecosystems.

Simulate biodiversity changes under climate scenarios.



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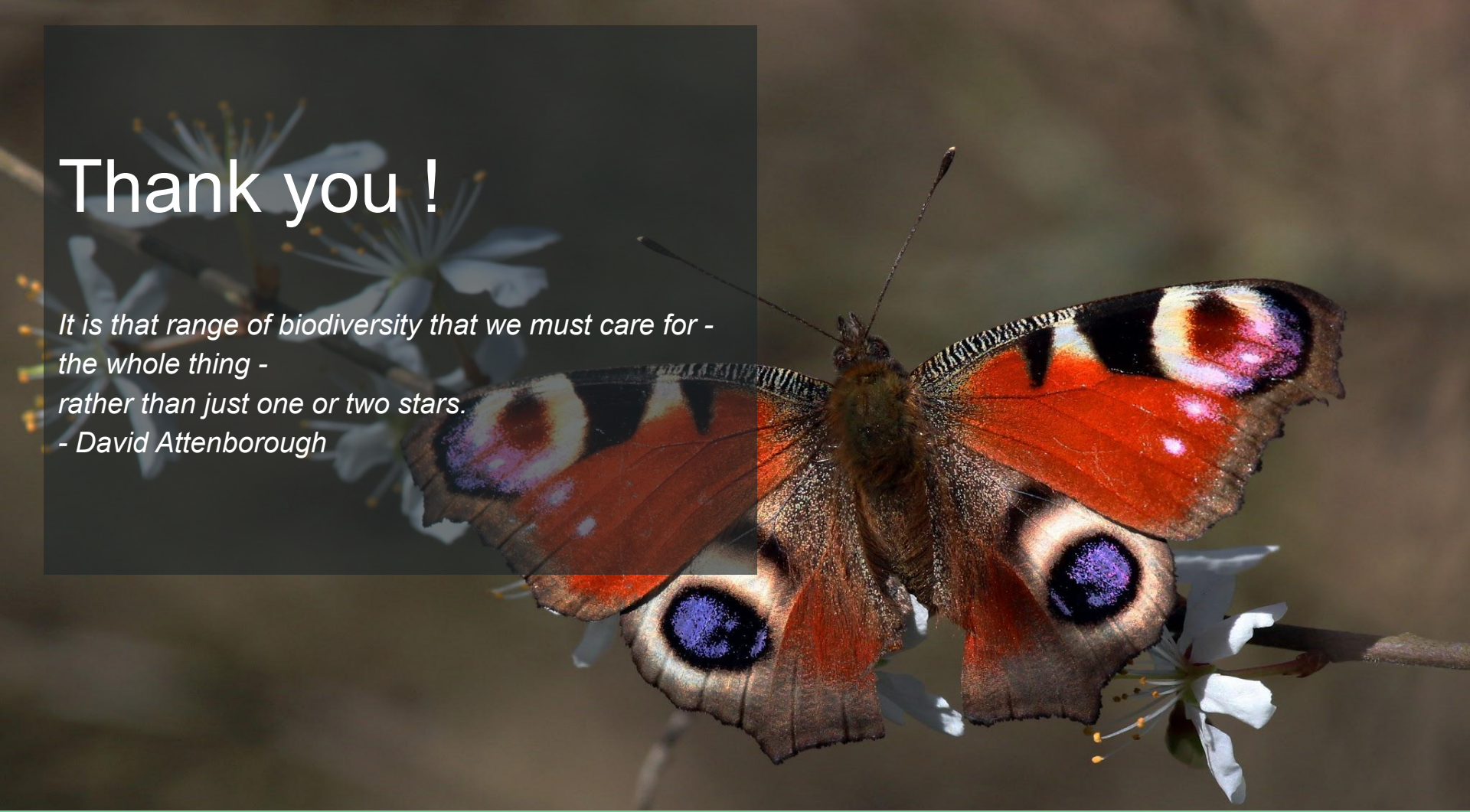
It is that range of biodiversity that we must care for -
the whole thing -
rather than just one or two stars.
- David Attenborough